

# OTACILIO B. L. NETO

## Postdoctoral Researcher

✉ otacilio.bneto@pm.me

🌐 tiominho.github.io

🔗 tiominho

🌐 tiominho

📄 0000-0001-5558-1242

## EDUCATION

D.Sc. (Tech.) in Chemical Engineering

**Aalto University**

📅 Nov 2021 – Dec 2025 🇫🇮 Finland

🏆 Recipient of the “Award for Excellence”

**Area:** Process control and automation

**Thesis:** Advances to predictive control and feed-back equilibrium seeking with applications to autonomous water resource recovery

**Supervisor(s):** Prof. Francesco Corona  
Prof. Iiro Harjunoski

M.Sc. in Teleinformatics Engineering

**Federal University of Ceará**

📅 Aug 2019 – Aug 2021 🇧🇷 Brazil

🏆 Awarded the “Academic Excellence Scholarship”

**Area:** Pattern recognition and dynamical systems

**Thesis:** Optimal control and estimation of activated sludge plants

B.Eng. in Computer Engineering

**Federal University of Ceará**

📅 Feb 2016 – Jul 2019 🇧🇷 Brazil

🏆 First class honours

**Thesis:** Optimal control: An application to a non-isothermal continuous reactor

## SKILLS

Optimal control Convex optimization

Multi-agent systems Game theory

Operator theory Numerical methods

State estimation Machine learning

Python (+ PyTorch, JAX, ...) C/C++

Julia MATLAB (+ CasADi) Linux

LaTeX Git

## LANGUAGES

Portuguese (Native) ● ● ● ● ●

English ● ● ● ● ●

Spanish ● ● ● ● ●

## EXPERIENCE

Postdoctoral Researcher

**Federal University of Ceará**

📅 Apr 2026 – present 🇧🇷 Fortaleza, Brazil

- Working on the distributed control of water treatment infrastructures
- Advising bachelor's and master's thesis on automatic control

Postdoctoral Researcher

**Aalto University**

📅 Jan 2026 – Mar 2026 🇫🇮 Espoo, Finland

- Responsible for graduate course “CHEM-E7225 Advanced Process Control”

Doctoral Researcher

**Aalto University**

📅 Nov 2021 – Dec 2025 🇫🇮 Espoo, Finland

- Worked on predictive control and equilibrium-seeking algorithms in the context of large-scale and multi-agent systems
- Teaching assistant for course “CHEM-E7225 Advanced Process Control”
- Teaching assistant for course “CHEM-E7190 Process Dynamics and Control”
- Advised a bachelor's thesis and a master's thesis

Undergraduate Researcher (Intern)

**Insight Data Science Lab**

📅 Feb 2018 – Jan 2019 🇧🇷 Fortaleza, Brazil

- Worked on ontological modelling, integration, and semantic compression of large-scale heterogeneous databases.

## PROJECTS

Control4Reuse

**Water JPI call “Process control technologies for water reuse”**

📅 Aug 2019 – Nov 2022 🇸🇪 Sweden + 🇧🇷 Brazil + 🇫🇷 France

- Researcher on the Work-Package 2 (Optimal Control and Estimation)

## SERVICES AND OTHER ACTIVITIES

**Peer-review for scientific publications**

- IEEE Transactions on Automatic Control 📅 2026 – present
- IEEE Control Systems Letters 📅 2025 – present
- Journal of Process Control 📅 2023 – present

**Peer-review for scientific events**

- Int. Symp. on Math. Theory of Networks and Systems 📅 2024
- American Control Conference 📅 2023
- IFAC World Congress 📅 2021 – 2023
- IFAC Symp. on Dynamics and Control of Process Systems 📅 2022
- IFAC Symp. on Advanced Control of Chemical Processes 📅 2021

**Organizing committees**

- 25th Nordic Process Control Workshop (NPCW) 📅 2025
- Ceará Informatics Olympics 📅 2016 – 2019

## REFERENCES

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Available under request.

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## PUBLICATION LIST

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Visit my **Google Scholar** for a more complete list.

### Journal Articles

- J. Magalhães, **O. Neto**, I. Harjunkski, and F. Corona, “Local particle flow filters for large-scale dynamic systems,” *Under review*, 2026.
- **O. Neto**, M. Mulas, and F. Corona, “A system level approach to generalized feedback Nash equilibrium seeking in partially-observed games,” *IEEE Transactions on Cybernetics (In Press)*, 2026.
- **O. Neto**, M. Mulas, I. Harjunkski, and F. Corona, “Predictive control of wastewater treatment plants as energy-autonomous water resource recovery facilities,” *Control Engineering Practice*, 2026.
- **O. Neto**, M. Mulas, and F. Corona, “SLS-BRD: A system-level approach to seeking generalized feedback Nash equilibria,” *IEEE Transactions on Automatic Control*, 2025.
- **O. Neto**, M. Mulas, and F. Corona, “A model-based framework for controlling activated sludge plants,” *Chemical Engineering Journal*, 2024.
- **O. Neto**, M. Mulas, and F. Corona, “About the classical and structural controllability and observability of a common class of activated sludge plants,” *Journal of Process Control*, 2022.

### Conference Proceedings

- J. Magalhães, **O. Neto**, and F. Corona, “Block particle filters for state estimation of stochastic reaction-diffusion systems,” in *22nd IFAC World Congress*, 2023.
- **O. Neto**, M. Mulas, and F. Corona, “Online optimal estimation and control for a common class of activated sludge plants,” in *13th DYCOPS*, 2022.
- **O. Neto**, A. Haddon, F. Aichouche, J. Harmand, M. Mulas, and F. Corona, “Predictive control of activated sludge plants to supply nitrogen for optimal crop growth,” in *11th ADCHEM*, 2021.
- **O. Neto**, M. Mulas, and F. Corona, “On the observability of activated sludge plants,” in *21th IFAC World Congress*, 2020.
- **O. Neto**, M. Mulas, and F. Corona, “On the controllability of activated sludge plants,” in *2020 ECC*, 2020.